Sources: Sheldon Axler, *Measure, Integration & Real Analysis* (MIRA) and its *Supplement*, at measure.axler.net.

Lemma 1. Suppose $a < b$ and $u\colon \mathbf{R} \to \mathbf{R}$ is Borel measurable with $u|_{[a,b]}$ continuous. Then $\int_a^b u(x)\,dx = \int \chi_{[a,b]} u\,d\lambda$, where the left side is the Riemann integral.

Proof. The restriction $u|_{[a,b]}$ is bounded, because a continuous function on a closed bounded set is bounded, and it is Riemann integrable because continuous functions are Riemann integrable (1.11). Thus the equality of the Riemann and Lebesgue integrals (3.34) gives $\int_a^b u(x)\,dx = \int_{[a,b]} u\,d\lambda$, and the right side equals $\int \chi_{[a,b]} u\,d\lambda$, because by integrals via finite simple functions (3.9), applied to u^+ and u^-, both are the supremum of the same sums $\sum_{j=1}^m c_j \lambda(A_j)$, the sets A_j with $c_j > 0$ lying in $[a, b]$ in both cases. □

Exercise 3.1 (MIRA 3B.14). Let λ denote Lebesgue measure on $\mathbf{R}$.

(a) Let $f(x) = 1/\sqrt{x}$. Prove that $\int_{[0,1]} f\,d\lambda = 2$.

(b) Let $f(x) = 1/(1+x^2)$. Prove that $\int_{\mathbf{R}} f\,d\lambda = \pi$.

(c) Let $f(x) = (\sin x)/x$. Show that the integral $\int_{(0,\infty)} f\,d\lambda$ is not defined but $\lim_{t\to\infty} \int_{(0,t)} f\,d\lambda$ exists in $\mathbf{R}$.

Solution 3.1. (a) Define $f\colon \mathbf{R} \to [0,\infty)$ by $f(x) = 1/\sqrt{x}$ for $x > 0$ and $f(x) = 0$ for $x \le 0$, which agrees on $[0,1]$ with the unbounded function of Example 1.15. Then f is Borel measurable, because $f|_{(0,\infty)}$ is continuous, every continuous function is Borel measurable (2.41), and f is 0 on $(-\infty, 0]$. By the definition of integration on a subset (3.24), $\int_{[0,1]} f\,d\lambda = \int \chi_{[0,1]} f\,d\lambda$. For $j \in \mathbf{Z}^+$ let $g_j = \chi_{[1/j,1]} f$, which is nonnegative and measurable. Because $f \ge 0$ and $[1/j, 1] \subseteq [1/(j+1), 1]$, we have $0 \le g_1 \le g_2 \le \cdots$, and $\lim_{j\to\infty} g_j(x) = \chi_{[0,1]}(x) f(x)$ for every $x \in \mathbf{R}$: if $x \in (0,1]$, then $g_j(x) = f(x)$ for all $j \ge 1/x$, and otherwise both sides are 0 because $f(0) = 0$ and $\chi_{[0,1]}(x) = 0$ for $x \notin [0,1]$. Thus the Monotone Convergence Theorem (3.11), Lemma 1 applied to f, which is continuous on $[1/j, 1]$, and the fundamental theorem of calculus give

$$\int_{[0,1]} f\,d\lambda = \lim_{j\to\infty} \int \chi_{[1/j,1]} f\,d\lambda = \lim_{j\to\infty} \int_{1/j}^1 \frac{dx}{\sqrt{x}} = \lim_{j\to\infty} \Big(2 - \frac{2}{\sqrt{j}}\Big) = 2.$$

(b) The function $f(x) = 1/(1+x^2)$ is continuous on $\mathbf{R}$, hence Borel measurable, and $f > 0$. By the definition of integration on a subset (3.24), $\int_{\mathbf{R}} f\,d\lambda = \int f\,d\lambda$. For $j \in \mathbf{Z}^+$ let $f_j = \chi_{[-j,j]} f$. Then $f_1, f_2, \ldots$ are nonnegative and measurable with $f_1 \le f_2 \le \cdots$, and $\lim_{j\to\infty} f_j(x) = f(x)$ for every $x \in \mathbf{R}$ because $x \in [-j, j]$ for all $j \ge |x|$. Thus the Monotone Convergence Theorem and Lemma 1 (f is continuous on $[-j, j]$) give

$$\int_{\mathbf{R}} f\,d\lambda = \lim_{j\to\infty} \int_{-j}^{j} \frac{dx}{1+x^2} = \lim_{j\to\infty} \big(\arctan j - \arctan(-j)\big) = \lim_{j\to\infty} 2\arctan j = \pi,$$

because $\arctan$ is an odd antiderivative of $x \mapsto 1/(1+x^2)$ and $\lim_{t\to\infty} \arctan t = \pi/2$.

(c) Now $f(x) = (\sin x)/x$ is continuous on $(0,\infty)$. Define $g\colon \mathbf{R} \to \mathbf{R}$ by $g(x) = f(x)$ for $x > 0$ and $g(x) = 0$ for $x \le 0$; then g is Borel measurable as in (a), and $|g| \le 1$ because $|\sin x| \le x$ for $x > 0$. Because g extends f by 0, the definition of integration on a subset (3.24) gives $\int_{(0,\infty)} f\,d\lambda = \int_{(0,\infty)} g\,d\lambda = \int \chi_{(0,\infty)} g\,d\lambda = \int g\,d\lambda$, each side defined exactly when the next is.

First we show that $\int g\,d\lambda$ is not defined. For $i \in \mathbf{Z}^+$ let $I_i = [i\pi, (i+1)\pi]$; on I_i the sign of $\sin x$ is that of $(-1)^i$, and $1/x \geq 1/((i+1)\pi)$. Suppose $n \in \mathbf{Z}^+$. The intervals $I_2, I_4, \ldots, I_{2n}$ are disjoint, and thus the measurable function $\varphi_n(x) = \sum_{m=1}^{n} \chi_{I_{2m}}(x)(\sin x)/\big((2m+1)\pi\big)$ satisfies $0 \leq \varphi_n \leq g^+$ on $\mathbf{R}$. The mth term is Borel measurable and continuous on I_{2m}, and thus Lemma 1 and the fundamental theorem of calculus give

$$\int \chi_{I_{2m}} \frac{\sin x}{(2m+1)\pi}\,d\lambda = \int_{2m\pi}^{(2m+1)\pi} \frac{\sin x}{(2m+1)\pi}\,dx = \frac{2}{(2m+1)\pi}.$$

Hence order preservation of integration (3.8) and additivity of integration for nonnegative functions (3.16) give

$$\int g^+\,d\lambda \geq \int \varphi_n\,d\lambda = \sum_{m=1}^{n} \frac{2}{(2m+1)\pi} \geq \frac{1}{\pi}\sum_{k=2}^{n+1} \frac{1}{k} = \frac{s_{n+1}-1}{\pi},$$

where $s_N = \sum_{k=1}^{N} 1/k$ and $2/(2m+1) \geq 1/(m+1)$. Because $s_{2^{M+1}} - s_{2^M}$ is a sum of 2^M terms, each at least 2^{-M-1}, we have $s_{2^M} \geq 1 + M/2$ for every integer $M \geq 0$. Taking $n = 2^M - 1$ gives $\int g^+\,d\lambda \geq M/(2\pi)$ for every $M \in \mathbf{Z}^+$; hence $\int g^+\,d\lambda = \infty$. In the same way, the intervals $I_1, I_3, \ldots, I_{2n-1}$ are disjoint, on I_{2m-1} we have $g^-(x) = -g(x) \geq (-\sin x)/(2m\pi) \geq 0$, and the measurable function $\psi_n(x) = \sum_{m=1}^{n} \chi_{I_{2m-1}}(x)(-\sin x)/(2m\pi)$ satisfies $0 \leq \psi_n \leq g^-$ on $\mathbf{R}$. As for g^+, Lemma 1 and the fundamental theorem of calculus give $\int \chi_{I_{2m-1}}(-\sin x)/(2m\pi)\,d\lambda = 2/(2m\pi)$. Thus order preservation (3.8) and additivity (3.16) of integration give $\int g^-\,d\lambda \geq \int \psi_n\,d\lambda = s_n/\pi$, which for $n = 2^M$ is at least $(1 + M/2)/\pi$. Thus $\int g^-\,d\lambda = \infty$ as well, and $\int g\,d\lambda$ is not defined by the definition of the integral of a real-valued function (3.18), which requires at least one of $\int g^+\,d\lambda$ and $\int g^-\,d\lambda$ to be finite; hence neither is $\int_{(0,\infty)} f\,d\lambda$.

Now let $F(t) = \int \chi_{(0,t)} g\,d\lambda$ for $t > 0$. If $A \subseteq [0,t]$ is a Borel set, then $|\chi_A g| \leq \chi_A$, and thus order preservation of integration (3.8) and the integral of a characteristic function (3.4) give $\int |\chi_A g|\,d\lambda \leq \lambda(A) \leq \lambda([0,t]) = t$; hence $\chi_A g \in \mathcal{L}^1(\lambda)$. In particular $F(t) \in \mathbf{R}$, and because $g|_{(0,t)} = f|_{(0,t)}$, the definition of integration on a subset (3.24) gives $\int_{(0,t)} f\,d\lambda = F(t)$.

Define $h, q\colon \mathbf{R} \to \mathbf{R}$ by $h(x) = (\cos x)/x^2$ and $q(x) = 1/x^2$ for $x \geq 1$ and $h(x) = q(x) = 0$ for $x < 1$. Both are Borel measurable as in (a), and $|h| \leq q$. Because $0 \leq \chi_{[1,2]} q \leq \chi_{[1,3]} q \leq \cdots$ and $\lim_{s\to\infty} \chi_{[1,s]} q = q$ pointwise, the Monotone Convergence Theorem and Lemma 1 (q is continuous on $[1,s]$) give

$$\int q\,d\lambda = \lim_{s\to\infty} \int_1^s \frac{dx}{x^2} = \lim_{s\to\infty} \left(1 - \frac{1}{s}\right) = 1,$$

and for $t > 1$ the same computation on $[t,s]$ gives $\int \chi_{[t,\infty)} q\,d\lambda = \lim_{s\to\infty}(1/t - 1/s) = 1/t$. Hence $\int |h|\,d\lambda \leq 1$ by order preservation of integration (3.8), and thus $h \in \mathcal{L}^1(\lambda)$ and $\chi_{(t,\infty)} h \in \mathcal{L}^1(\lambda)$; let $H = \int h\,d\lambda$. Thus, because the absolute value of an integral is at most the integral of the absolute value (3.23) and integration preserves order (3.8), we have $\left|\int \chi_{(t,\infty)} h\,d\lambda\right| \leq \int \chi_{(t,\infty)} |h|\,d\lambda \leq \int \chi_{[t,\infty)} q\,d\lambda = 1/t$.

Suppose $t > 1$. The function $u(x) = -(\cos x)/x$ satisfies $u'(x) = (\sin x)/x + (\cos x)/x^2 = g(x) + h(x)$ on $[1,t]$, where g and h are continuous. Thus the fundamental theorem of calculus and Lemma 1 give

$$\int \chi_{[1,t]} g\,d\lambda = \int_1^t g(x)\,dx = u(t) - u(1) - \int_1^t h(x)\,dx = \cos 1 - \frac{\cos t}{t} - \int \chi_{[1,t]} h\,d\lambda.$$

Now $\chi_{(0,t)} g + \chi_{\{t\}} g = \chi_{(0,1)} g + \chi_{[1,t]} g$ and $h = \chi_{[1,t]} h + \chi_{(t,\infty)} h$ on $\mathbf{R}$. Each function in these two equations is in $\mathcal{L}^1(\lambda)$ by the two bounds above, and $\int \chi_{\{t\}} g\,d\lambda = 0$ because $\lambda(\{t\}) = 0$. Thus

additivity of integration (3.21), applied to both equations, gives

$$F(t) = c + \int \chi_{[1,t]} g \, d\lambda = c + \cos 1 - \frac{\cos t}{t} - H + \int \chi_{(t,\infty)} h \, d\lambda,$$

where $c = \int \chi_{(0,1)} g \, d\lambda \in \mathbf{R}$. Let $L = c + \cos 1 - H$. Because $|\cos t| \le 1$ and $\left|\int \chi_{(t,\infty)} h \, d\lambda\right| \le 1/t$, the triangle inequality gives $|F(t) - L| \le 2/t$ for all $t > 1$. Hence $\lim_{t\to\infty} \int_{(0,t)} f \, d\lambda = L$ exists in $\mathbf{R}$, as desired. □

Lemma 2. Suppose $h \colon \mathbf{R} \to [0,\infty)$ is Borel measurable, $a \in \mathbf{R}$, $s \in \mathbf{R} \setminus \{0\}$, $h_a(x) = h(x-a)$ and $h^{(s)}(x) = h(sx)$. Then

$$\int h_a \, d\lambda = \int h \, d\lambda \qquad \text{and} \qquad \int h^{(s)} \, d\lambda = \frac{1}{|s|} \int h \, d\lambda.$$

Moreover, $x \mapsto u(x-a)$ and $x \mapsto u(sx)$ are Borel measurable for every Borel measurable $u \colon \mathbf{R} \to \mathbf{R}$.

Proof. The maps $x \mapsto x - a$ and $x \mapsto sx$ are continuous, and thus by the composition of measurable functions (2.44) the functions $u(x-a)$ and $u(sx)$ are Borel measurable for every Borel measurable u, and $a + A$ and $s^{-1}A$ are Borel sets for every Borel set A. If $A_1, \dots, A_m$ is a $\mathcal{B}$-partition of $\mathbf{R}$, then so is $a + A_1, \dots, a + A_m$, with $\inf_{a+A_j} h_a = \inf_{A_j} h$ and $\lambda(a + A_j) = \lambda(A_j)$ because outer measure is translation invariant (2.7); every $\mathcal{B}$-partition of $\mathbf{R}$ is such a translate, and thus the lower Lebesgue sums (3.2) of h_a and of h have the same supremum, which is the first equation. Scaling by s the open intervals of a cover of A gives a cover of sA with lengths multiplied by $|s|$, and thus $|sA| \le |s|\,|A|$ by the definition of outer measure (2.2); applying this to sA and $1/s$ gives $\lambda(sA) = |s|\,\lambda(A)$, and the partitions $s^{-1}A_1, \dots, s^{-1}A_m$ give the second equation in the same way. □

Exercise 3.2 (MIRA 3B.16). Suppose $f \in \mathcal{L}^1(\mathbf{R})$.

(a) For $t \in \mathbf{R}$, define $f_t \colon \mathbf{R} \to \mathbf{R}$ by $f_t(x) = f(x-t)$. Prove that

$$\lim_{t\to 0} \|f - f_t\|_1 = 0.$$

(b) For $t > 0$, define $f_t \colon \mathbf{R} \to \mathbf{R}$ by $f_t(x) = f(tx)$. Prove that

$$\lim_{t\to 1} \|f - f_t\|_1 = 0.$$

Solution 3.2. Here λ is Lebesgue measure on the Borel subsets of $\mathbf{R}$, as in the definition of $\mathcal{L}^1(\mathbf{R})$ (3.45); thus f is Borel measurable and $\|f\|_1 = \int |f| \, d\lambda < \infty$.

(a) Suppose $\varepsilon > 0$. Approximation by continuous functions (3.48) gives a continuous $g \colon \mathbf{R} \to \mathbf{R}$ with $\|f - g\|_1 < \varepsilon/3$ and $\{x \in \mathbf{R} : g(x) \ne 0\}$ bounded, hence contained in $[-M, M]$ for some $M > 0$. Because g is continuous, $g - f$ is Borel measurable with $\|g - f\|_1 < \varepsilon/3$, and thus $g = f + (g - f) \in \mathcal{L}^1(\mathbf{R})$ by the triangle inequality for the $\mathcal{L}^1$-norm (the fourth bullet of 3.43). Let $\eta = \varepsilon/\big(3(2M+2)\big)$. Because g is continuous on the closed bounded set $[-M-1, M+1]$ and continuous functions on closed bounded sets are uniformly continuous, there exists $\delta \in (0, 1]$ such that $|g(x) - g(y)| < \eta$ for all $x, y \in [-M-1, M+1]$ with $|x - y| < \delta$. If $x, y \in \mathbf{R}$ with $|x - y| < \delta$ do not both lie in $[-M-1, M+1]$, say $|x| > M + 1$, then $|y| \ge |x| - |x - y| > M$ by the triangle inequality, which implies that $g(x) = g(y) = 0$. Hence $|g(x) - g(y)| < \eta$ for all $x, y \in \mathbf{R}$ with $|x - y| < \delta$.

Now suppose $t \in \mathbf{R}$ with $|t| < \delta$. Taking $y = x - t$ above gives $|g(x) - g_t(x)| < \eta$ for all $x \in \mathbf{R}$, and $g(x) = g_t(x) = 0$ when $|x| > M + 1$. Hence $|g - g_t| \le \eta\, \chi_{[-M-1,M+1]}$. Thus order preservation of integration (3.8) and the integral of a simple function (3.7) give $\|g - g_t\|_1 \le \eta\, \lambda([-M-1, M+1]) = \eta(2M+2) = \varepsilon/3$. Next, $|f_t - g_t|$, $|f_t|$ and $|g_t|$ are the translates by t of the nonnegative functions $|f-g|$, $|f|$ and $|g|$. Thus the first equation of Lemma 2 gives $\|f_t - g_t\|_1 = \|f - g\|_1 < \varepsilon/3$, $\|f_t\|_1 = \|f\|_1 < \infty$ and $\|g_t\|_1 = \|g\|_1 < \infty$; in particular, $f_t, g_t \in \mathcal{L}^1(\mathbf{R})$. Now $f - f_t = (f-g) + (g - g_t) + (g_t - f_t)$, and thus the triangle inequality for the $\mathcal{L}^1$-norm gives

$$\|f - f_t\|_1 \le \|f - g\|_1 + \|g - g_t\|_1 + \|f_t - g_t\|_1 < \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon$$

for all $t \in \mathbf{R}$ with $|t| < \delta$. Thus $\lim_{t\to 0} \|f - f_t\|_1 = 0$.

(b) Suppose $\varepsilon > 0$. By approximation by continuous functions (3.48), there exists a continuous $g\colon \mathbf{R} \to \mathbf{R}$ with $\|f - g\|_1 < \varepsilon/6$ and $\{x \in \mathbf{R} : g(x) \ne 0\} \subseteq [-M, M]$ for some $M > 0$. As in (a), $g \in \mathcal{L}^1(\mathbf{R})$. Let $\eta = \varepsilon/(12M)$. Because g is continuous on the closed bounded set $[-3M, 3M]$ and continuous functions on closed bounded sets are uniformly continuous, there exists $\delta > 0$ such that $|g(x) - g(y)| < \eta$ for all $x, y \in [-3M, 3M]$ with $|x - y| < \delta$. Let $\rho = \min\{\frac{1}{2}, \delta/(2M)\}$, and suppose $t > 0$ with $|t - 1| < \rho$; then $\frac{1}{2} < t < \frac{3}{2}$. If $x \in \mathbf{R}$ and $|x| \le 2M$, then $x, tx \in [-3M, 3M]$ because $t < \frac{3}{2}$, and $|x - tx| = |x|\,|1 - t| < 2M\rho \le \delta$, and thus $|g(x) - g_t(x)| < \eta$; if $|x| > 2M$, then $|x| > M$ and $|tx| > M$, and thus $g(x) = g_t(x) = 0$. Hence $|g - g_t| \le \eta\, \chi_{[-2M,2M]}$. Thus order preservation (3.8) and the integral of a simple function (3.7) give $\|g - g_t\|_1 \le \eta\, \lambda([-2M, 2M]) = 4M\eta = \varepsilon/3$. Because $|f_t - g_t| = |f - g|^{(t)}$ and $1/t < 2$, the second equation of Lemma 2, applied to the nonnegative function $|f - g|$, gives $\|f_t - g_t\|_1 = \frac{1}{t}\|f - g\|_1 < 2 \cdot \frac{\varepsilon}{6} = \frac{\varepsilon}{3}$; applied to $|f|$ and $|g|$ it gives $\|f_t\|_1 = \|f\|_1/t < \infty$ and $\|g_t\|_1 = \|g\|_1/t < \infty$, and thus $f_t, g_t \in \mathcal{L}^1(\mathbf{R})$. Thus the triangle inequality for the $\mathcal{L}^1$-norm gives

$$\|f - f_t\|_1 \le \|f - g\|_1 + \|g - g_t\|_1 + \|f_t - g_t\|_1 < \frac{\varepsilon}{6} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} < \varepsilon$$

for all $t > 0$ with $|t - 1| < \rho$. Hence $\lim_{t\to 1} \|f - f_t\|_1 = 0$. □

Exercise 3.3 (MIRA 4A.1). Suppose $(X, \mathcal{S}, \mu)$ is a measure space and $h\colon X \to \mathbf{R}$ is an $\mathcal{S}$-measurable function. Prove that

$$\mu(\{x \in X : |h(x)| \ge c\}) \le \frac{1}{c^p} \int |h|^p \, d\mu$$

for all positive numbers c and p.

Solution 3.3. Suppose $c > 0$ and $p > 0$, and let $A = \{x \in X : |h(x)| \ge c\}$. The function $|h|^p$ is $\mathcal{S}$-measurable by composition of measurable functions (2.44), being the composition of h with the continuous function $y \mapsto |y|^p$. Because $y \mapsto y^p$ is strictly increasing on $[0, \infty)$, we have $A = \{x \in X : |h(x)|^p \ge c^p\}$, the inverse image of the closed set $[c^p, \infty)$ under $|h|^p$. Thus $A \in \mathcal{S}$ and $c^p \chi_A \le |h|^p$ on X. Hence the integral of a simple function (3.7) and order preservation of integration (3.8) give

$$c^p \mu(A) = \int c^p \chi_A \, d\mu \le \int |h|^p \, d\mu.$$

Dividing by $c^p > 0$ gives the desired inequality. □

Exercise 3.4 (MIRA 4A.2). Suppose $(X, \mathcal{S}, \mu)$ is a measure space with $\mu(X) = 1$ and $h \in \mathcal{L}^1(\mu)$. Prove that

$$\mu\Big(\Big\{x \in X : \Big|h(x) - \int h\,d\mu\Big| \geq c\Big\}\Big) \leq \frac{1}{c^2}\Big(\int h^2\,d\mu - \Big(\int h\,d\mu\Big)^2\Big)$$

for all $c > 0$. [The result above is called Chebyshev's inequality; it plays an important role in probability theory. Pafnuty Chebyshev (1821–1894) was Markov's thesis advisor.]

Solution 3.4. Let $a = \int h\,d\mu$, which is a real number because $h \in \mathcal{L}^1(\mu)$, and suppose $c > 0$. The function h^2 is $\mathcal{S}$-measurable. If $\int h^2\,d\mu = \infty$, then the right side of the desired inequality is ∞ and the inequality holds. Thus assume $\int h^2\,d\mu < \infty$.

Let $k = h - a$, which is $\mathcal{S}$-measurable because a difference of $\mathcal{S}$-measurable functions is $\mathcal{S}$-measurable (2.46(a)). Now $k^2 = (h^2 + a^2) - 2ah$ on X. Additivity of integration for nonnegative functions (3.16) and the integral of a simple function (3.7) give $\int (h^2 + a^2)\,d\mu = \int h^2\,d\mu + a^2\mu(X) = \int h^2\,d\mu + a^2 < \infty$, because $\mu(X) = 1$. Homogeneity of integration (3.20) gives $\int |2ah|\,d\mu = 2|a|\|h\|_1 < \infty$. Thus additivity of integration (3.21) and homogeneity (3.20) give $\int k^2\,d\mu = \int h^2\,d\mu + a^2 - 2a\int h\,d\mu = \int h^2\,d\mu - a^2$. Hence Exercise 4A.1 applied to k with $p = 2$ gives

$$\begin{aligned}\mu\Big(\Big\{x \in X : \Big|h(x) - \int h\,d\mu\Big| \geq c\Big\}\Big) &= \mu(\{x \in X : |k(x)| \geq c\}) \\ &\leq \frac{1}{c^2}\int k^2\,d\mu = \frac{1}{c^2}\Big(\int h^2\,d\mu - \Big(\int h\,d\mu\Big)^2\Big). \qquad \square\end{aligned}$$

Exercise 3.5 (MIRA 4A.10). Prove or give a counterexample: If $h\colon \mathbf{R} \to [0, \infty)$ is an increasing function, then h^* is an increasing function.

Solution 3.5. The statement is true. Suppose $h\colon \mathbf{R} \to [0, \infty)$ is increasing. Then h is Borel measurable because every increasing function is Borel measurable (2.43), and $|h| = h$. We take λ to be Lebesgue measure on the Borel sets, as in the definition of Lebesgue measure (2.69) and in Exercise 3B.16.

Suppose $b < b'$, let $s = b' - b > 0$, and suppose $t > 0$. Let $f = \chi_{(b'-t,b'+t)}h$ and $f' = \chi_{(b-t,b+t)}h$, nonnegative Borel measurable functions with $\int f\,d\lambda = \int_{b'-t}^{b'+t} h$ and $\int f'\,d\lambda = \int_{b-t}^{b+t} h$ by the definition of $\int_a^b$ (3.39). Let $g(x) = f(x + s)$ for $x \in \mathbf{R}$, also nonnegative and Borel measurable. The first equation of Lemma 2, with $a = -s$, gives $\int g\,d\lambda = \int f\,d\lambda$. If $x \in \mathbf{R}$, then $x + s \in (b' - t, b' + t)$ if and only if $x \in (b - t, b + t)$, and $h(x) \leq h(x + s)$ because h is increasing; thus $f'(x) = \chi_{(b-t,b+t)}(x)h(x) \leq \chi_{(b-t,b+t)}(x)h(x + s) = g(x)$. Hence order preservation of integration (3.8) gives

$$\frac{1}{2t}\int_{b-t}^{b+t} h = \frac{1}{2t}\int f'\,d\lambda \leq \frac{1}{2t}\int g\,d\lambda = \frac{1}{2t}\int f\,d\lambda = \frac{1}{2t}\int_{b'-t}^{b'+t} h \leq h^*(b').$$

Taking the supremum over $t > 0$ gives $h^*(b) \leq h^*(b')$ by the definition of the Hardy–Littlewood maximal function (4.6). Thus h^* is increasing. $\square$

For $f \in \mathcal{L}^1(\mathbf{R})$ and I an interval of $\mathbf{R}$ with $0 < |I| < \infty$, let f_I denote the average of f on I. In other words, $f_I = \frac{1}{|I|}\int_I f$.

Lemma 3. Suppose $u < v$ and I is an interval with $(u, v) \subseteq I \subseteq [u, v]$. Then $|I| = v - u$, and $\int_I g = \int_u^v g$ for every Lebesgue measurable $g\colon \mathbf{R} \to [0, \infty)$. If $f \in \mathcal{L}^1(\mathbf{R})$ and $b \in \mathbf{R}$, then $\int_I |f - f_I| \le 2 \int_I |f - f(b)|$.

Proof. The set I is a Borel set, being the union of (u, v) and a subset of $\{u, v\}$. Because outer measure preserves order (2.5) and the outer measure of a closed interval is its length (2.14), we have $|I| \le v - u$. Finite sets have outer measure 0 (Example 2.3), and thus subadditivity of outer measure (2.8) gives $v - u = |[u, v]| \le |I| + |\{u, v\}| = |I|$. Hence $|I| = v - u$. Now suppose $g\colon \mathbf{R} \to [0, \infty)$ is Lebesgue measurable, and let $E = I \setminus (u, v)$, a subset of $\{u, v\}$. Then $|E| = 0$ and $g \le M$ on E, where $M = \max\{g(u), g(v)\}$. Order preservation of integration (3.8) and the integral of a simple function (3.7) give $0 \le \int_E g \le M|E| = 0$. Because $\chi_I g = \chi_{(u,v)} g + \chi_E g$, additivity of integration for nonnegative functions (3.16) and the definition of $\int_a^b$ (3.39) give $\int_I g = \int_{(u,v)} g + \int_E g = \int_u^v g$.

Now suppose $f \in \mathcal{L}^1(\mathbf{R})$ and $b \in \mathbf{R}$. Because $|\chi_I f| \le |f|$, order preservation of integration (3.8) gives $\chi_I f \in \mathcal{L}^1(\mathbf{R})$, and thus f_I is defined. By the integral of a simple function (3.7) and homogeneity of integration (3.20), $\int_I f(b) = f(b)|I|$. Thus additivity of integration (3.21) and homogeneity (3.20) give $\int_I \big(f(b) - f\big) = |I|\big(f(b) - f_I\big)$. Because the absolute value of an integral is at most the integral of the absolute value (3.23), we have $|I||f(b) - f_I| \le \int_I |f - f(b)|$. By the triangle inequality, $|f - f_I| \le |f - f(b)| + |f(b) - f_I|$ on $\mathbf{R}$. Multiplying by χ_I and integrating, order preservation (3.8) and additivity (3.16) of integration give $\int_I |f - f_I| \le \int_I |f - f(b)| + |f(b) - f_I||I| \le 2 \int_I |f - f(b)|$, where the last step is the inequality above. □

Exercise 3.6 (MIRA 4B.1). Suppose $f \in \mathcal{L}^1(\mathbf{R})$. Prove that

$$\lim_{t \downarrow 0} \frac{1}{2t} \int_{b-t}^{b+t} |f - f_{[b-t,b+t]}| = 0$$

for almost every $b \in \mathbf{R}$.

Solution 3.6. Suppose $b \in \mathbf{R}$ and $t > 0$, and let $I = [b - t, b + t]$; then $|I| = 2t$ by Lemma 3, and

$$0 \le \frac{1}{2t} \int_{b-t}^{b+t} |f - f_{[b-t,b+t]}| = \frac{1}{2t} \int_I |f - f_I| \le \frac{1}{t} \int_I |f - f(b)| = 2 \cdot \frac{1}{2t} \int_{b-t}^{b+t} |f - f(b)|,$$

where both equations and the inequality between them come from Lemma 3. The right side tends to 0 as $t \downarrow 0$ for almost every $b \in \mathbf{R}$ by the first version of the Lebesgue Differentiation Theorem (4.10), completing the proof. □

Exercise 3.7 (MIRA 4B.2). Suppose $f \in \mathcal{L}^1(\mathbf{R})$. Prove that

$$\lim_{t \downarrow 0} \sup \left\{ \frac{1}{|I|} \int_I |f - f_I| : I \text{ is an interval of length } t \text{ containing } b \right\} = 0$$

for almost every $b \in \mathbf{R}$.

Solution 3.7. Suppose $b \in \mathbf{R}$, $t > 0$, and I is an interval of length t containing b. Because $0 < |I| < \infty$, the interval I is bounded and contains more than one point, and thus has endpoints $u < v$ in $\mathbf{R}$ with $(u, v) \subseteq I \subseteq [u, v]$; by Lemma 3, $v - u = |I| = t$. Because $u \le b \le v$, we have $u \ge b - t$ and $v \le b + t$; thus $I \subseteq [b - t, b + t]$ and $\chi_I |f - f(b)| \le \chi_{[b-t,b+t]} |f - f(b)|$ on $\mathbf{R}$.

By Lemma 3, $\int_I |f - f_I| \le 2 \int_I |f - f(b)|$ and $\int_{[b-t,b+t]} |f - f(b)| = \int_{b-t}^{b+t} |f - f(b)|$. Thus order preservation of integration (3.8) gives

$$0 \le \frac{1}{|I|} \int_I |f - f_I| \le \frac{2}{t} \int_{[b-t,b+t]} |f - f(b)| = 4 \cdot \frac{1}{2t} \int_{b-t}^{b+t} |f - f(b)|.$$

The right side does not depend on I. Hence the supremum in the statement lies between 0 and the right side, which tends to 0 as $t \downarrow 0$ for almost every $b \in \mathbf{R}$ by the first version of the Lebesgue Differentiation Theorem (4.10), as desired. □

Exercise 3.8 (MIRA 4B.3). Suppose $f : \mathbf{R} \to \mathbf{R}$ is a Lebesgue measurable function such that $f^2 \in \mathcal{L}^1(\mathbf{R})$. Prove that

$$\lim_{t \downarrow 0} \frac{1}{2t} \int_{b-t}^{b+t} |f - f(b)|^2 = 0$$

for almost every $b \in \mathbf{R}$.

Solution 3.8. For $n \in \mathbf{Z}^+$, let $f_n = \chi_{(-n,n)} f$, which is Lebesgue measurable. Because $0 \le (|f| - 1)^2$, we have $|f| \le \frac{1}{2}(1 + f^2)$ on $\mathbf{R}$, and thus $|f_n| \le \frac{1}{2}\chi_{(-n,n)} + \frac{1}{2} f^2$. Order preservation of integration (3.8) and additivity of integration for nonnegative functions (3.16) give $\int |f_n| \, d\lambda \le \int \frac{1}{2}\chi_{(-n,n)} \, d\lambda + \int \frac{1}{2} f^2 \, d\lambda$. The first integral on the right is $\frac{1}{2}|(-n,n)| = n$ by the integral of a simple function (3.7), and the second is $\frac{1}{2} \int f^2 \, d\lambda < \infty$ by homogeneity of integration (3.20). Thus $f_n \in \mathcal{L}^1(\mathbf{R})$. Because an $\mathcal{L}^1(\mathbf{R})$ function equals its local average almost everywhere (4.21), there are sets $A_0, A_1, A_2, \ldots$ with $|A_n| = 0$ such that $f(b)^2 = \lim_{t \downarrow 0} \frac{1}{2t} \int_{b-t}^{b+t} f^2$ for every $b \in \mathbf{R} \setminus A_0$ and $f_n(b) = \lim_{t \downarrow 0} \frac{1}{2t} \int_{b-t}^{b+t} f_n$ for every $b \in \mathbf{R} \setminus A_n$. Let $A = \bigcup_{n=0}^{\infty} A_n$; then $|A| = 0$ by countable subadditivity of outer measure (2.8).

Suppose $b \in \mathbf{R} \setminus A$. Let $n \in \mathbf{Z}^+$ be such that $n > |b|$; then $b \in (-n, n)$ and $f_n(b) = f(b)$. Suppose $0 < t < n - |b|$, and let $J = (b - t, b + t)$. Then $J \subseteq (-n, n)$, which implies that $\chi_J f_n = \chi_J f$, and $|J| = 2t$. On $\mathbf{R}$ we have $|f - f(b)|^2 = f^2 - 2f(b)f + f(b)^2$, and thus

$$\chi_J |f - f(b)|^2 = \chi_J f^2 - 2f(b)\, \chi_J f_n + f(b)^2 \chi_J.$$

The three functions on the right are Lebesgue measurable and in $\mathcal{L}^1(\mathbf{R})$: $|\chi_J f^2| \le f^2$ and $|\chi_J f_n| \le |f_n|$ on $\mathbf{R}$, which gives finite integrals by order preservation of integration (3.8), and $\int f(b)^2 \chi_J \, d\lambda = 2t f(b)^2$ by the integral of a simple function (3.7). Hence additivity of integration (3.21) and homogeneity (3.20) give

$$\int \chi_J |f - f(b)|^2 \, d\lambda = \int \chi_J f^2 \, d\lambda - 2f(b) \int \chi_J f_n \, d\lambda + 2t f(b)^2.$$

Dividing by $2t$ and using the definition of $\int_a^b$ (3.39), we have

$$\frac{1}{2t} \int_{b-t}^{b+t} |f - f(b)|^2 = \frac{1}{2t} \int_{b-t}^{b+t} f^2 - 2f(b) \cdot \frac{1}{2t} \int_{b-t}^{b+t} f_n + f(b)^2$$

for every $t \in (0, n - |b|)$. As $t \downarrow 0$, the first term on the right tends to $f(b)^2$ because $b \notin A_0$, and the second tends to $-2f(b)f_n(b) = -2f(b)^2$ because $b \notin A_n$. Thus the right side tends to $f(b)^2 - 2f(b)^2 + f(b)^2 = 0$. Because $|A| = 0$, this proves the desired limit for almost every $b \in \mathbf{R}$. □

Exercise 3.9 (MIRA 4B.4). Prove that the Lebesgue Differentiation Theorem (4.19) still holds if the hypothesis that $\int_{-\infty}^{\infty}|f|<\infty$ is weakened to the requirement that $\int_{-\infty}^{x}|f|<\infty$ for all $x\in\mathbf{R}$.

Solution 3.9. Suppose $f\colon\mathbf{R}\to\mathbf{R}$ is Lebesgue measurable and $\int_{-\infty}^{x}|f|<\infty$ for every $x\in\mathbf{R}$. For $x\in\mathbf{R}$, the function $\chi_{(-\infty,x)}f$ is Lebesgue measurable and $\int|\chi_{(-\infty,x)}f|\,d\lambda=\int_{-\infty}^{x}|f|<\infty$ by the definition of $\int_a^b$ (3.39); thus $\chi_{(-\infty,x)}f\in\mathcal{L}^1(\mathbf{R})$, and $g(x)=\int_{-\infty}^{x}f$ defines a function $g\colon\mathbf{R}\to\mathbf{R}$. For $n\in\mathbf{Z}^+$, let $f_n=\chi_{(-\infty,n)}f$, which is in $\mathcal{L}^1(\mathbf{R})$ by the case $x=n$, and define $g_n\colon\mathbf{R}\to\mathbf{R}$ by $g_n(x)=\int_{-\infty}^{x}f_n$. If $x\le n$, then $\chi_{(-\infty,x)}f_n=\chi_{(-\infty,x)}f$, and thus $g_n(x)=g(x)$. By the second version of the Lebesgue Differentiation Theorem (4.19), applied to f_n, there is a set A_n with $|A_n|=0$ such that $g_n'(b)=f_n(b)$ for every $b\in\mathbf{R}\setminus A_n$. Countable subadditivity of outer measure (2.8) gives $|A|=0$ for $A=\bigcup_{n=1}^{\infty}A_n$.

Now suppose $b\in\mathbf{R}\setminus A$. Let $n\in\mathbf{Z}^+$ be such that $n>b$. Then $g=g_n$ on the open interval $(-\infty,n)$, which contains b, and thus g and g_n have the same difference quotients $\big(g(b+t)-g(b)\big)/t$ for all t with $0<|t|<n-b$; hence by the definition of the derivative (4.16), g is differentiable at b if and only if g_n is, with the same derivative. Because $b\notin A_n$, g_n is differentiable at b with $g_n'(b)=f_n(b)$, and $f_n(b)=f(b)$ because $b<n$. Hence g is differentiable at b with $g'(b)=f(b)$ for every $b\in\mathbf{R}\setminus A$, and thus for almost every $b\in\mathbf{R}$, as desired. □

Exercise 3.10 (MIRA 4B.5). Suppose $f\colon\mathbf{R}\to\mathbf{R}$ is a Lebesgue measurable function. Prove that

$$|f(b)|\le f^*(b)$$

for almost every $b\in\mathbf{R}$.

Solution 3.10. For $n\in\mathbf{Z}^+$, let $E_n=\{x\in(-n,n):|f(x)|\le n\}$, a Lebesgue measurable set, and let $f_n=\chi_{E_n}f$, which is Lebesgue measurable, with $|f_n|\le n\chi_{(-n,n)}$ on $\mathbf{R}$. Thus order preservation of integration (3.8) and the integral of a simple function (3.7) give $\int|f_n|\,d\lambda\le n|(-n,n)|=2n^2<\infty$; hence $f_n\in\mathcal{L}^1(\mathbf{R})$. Because an $\mathcal{L}^1(\mathbf{R})$ function equals its local average almost everywhere (4.21), there is a set A_n with $|A_n|=0$ such that $f_n(b)=\lim_{t\downarrow0}\frac{1}{2t}\int_{b-t}^{b+t}f_n$ for every $b\in\mathbf{R}\setminus A_n$. Let $A=A_1\cup A_2\cup\cdots$. Then $|A|\le\sum_{n=1}^{\infty}|A_n|=0$ by countable subadditivity of outer measure (2.8).

Suppose $b\in\mathbf{R}\setminus A$. There exists $n\in\mathbf{Z}^+$ such that $n>|b|$ and $n>|f(b)|$. Then $b\in E_n$, and thus $f_n(b)=f(b)$. Suppose $t>0$. Because $f_n\in\mathcal{L}^1(\mathbf{R})$, the absolute value of an integral is at most the integral of the absolute value (3.23), and $|f_n|\le|f|$ on $\mathbf{R}$. Thus order preservation of integration (3.8) and the definition of f^* (4.6) give

$$\Big|\frac{1}{2t}\int_{b-t}^{b+t}f_n\Big|\le\frac{1}{2t}\int_{b-t}^{b+t}|f_n|\le\frac{1}{2t}\int_{b-t}^{b+t}|f|\le f^*(b).$$

Because $b\notin A_n$, letting $t\downarrow0$ gives $|f(b)|=|f_n(b)|\le f^*(b)$. This holds for every $b\in\mathbf{R}\setminus A$, and thus for almost every $b\in\mathbf{R}$. □